

Heart Disease Prediction Using Machine Learning Algorithms

Abstract

Heart disease is one of the leading causes of death worldwide. Early diagnosis can help reduce mortality rates and improve patient outcomes. This project uses machine learning techniques to predict the likelihood of heart disease based on patient medical data.

Introduction

Machine Learning is widely used in healthcare to assist in diagnosis and decision-making. By analyzing patient information, machine learning models can identify patterns associated with heart disease.

Objective

The objective of this project is to develop a machine learning-based system capable of predicting heart disease using patient health parameters.

Dataset Description

The dataset contains patient records with medical information such as:

- Age
- Sex
- Chest Pain Type
- Cholesterol Level
- Resting Blood Pressure
- Maximum Heart Rate
- Exercise Induced Angina

The target variable indicates whether heart disease is present or absent.

Methodology

Data Preprocessing

- Dataset loaded using Pandas
- Features and target separated

- Train-test split performed
- Data normalized using StandardScaler

Models Used

1. Logistic Regression
2. Random Forest Classifier

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve
- ROC-AUC Score

Results

The models were trained and evaluated on test data. Random Forest achieved the highest prediction accuracy and ROC-AUC score. The confusion matrix and classification report confirmed effective classification performance.

Classification Report:													
				precision		recall		f1-score		support			
			0	0.97		1.00		0.99		102			
			1	1.00		0.97		0.99		103			
			accuracy					0.99		205			
			macro avg	0.99		0.99		0.99		205			
			weighted avg	0.99		0.99		0.99		205			

... Dataset Shape:

(1025, 14)

First Five Rows:

	age	sex	cp	trestbps	chol	fbs	...	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	...	0	1.0	2	2	3	0
1	53	1	0	140	203	1	...	1	3.1	0	0	3	0
2	70	1	0	145	174	0	...	1	2.6	0	0	3	0
3	61	1	0	148	203	0	...	0	0.0	2	1	3	0
4	62	0	0	138	294	1	...	0	1.9	1	3	2	0

[5 rows x 14 columns]

... Logistic Regression Accuracy:

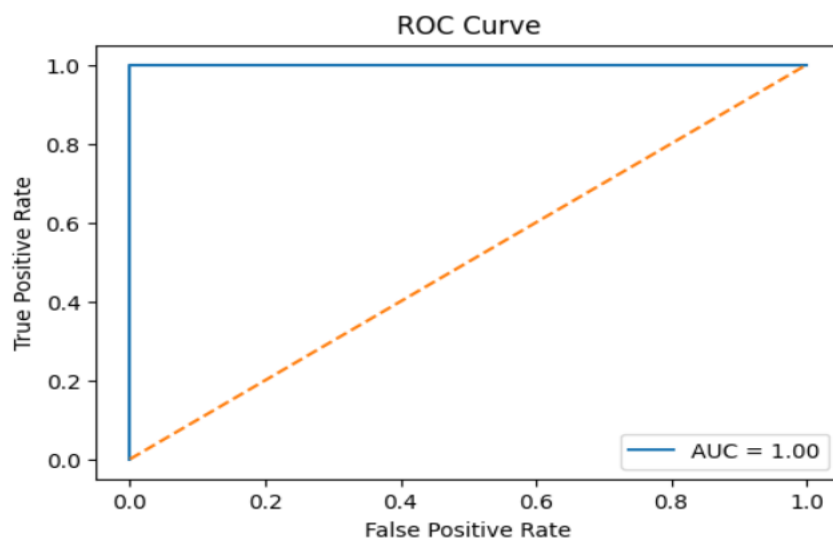
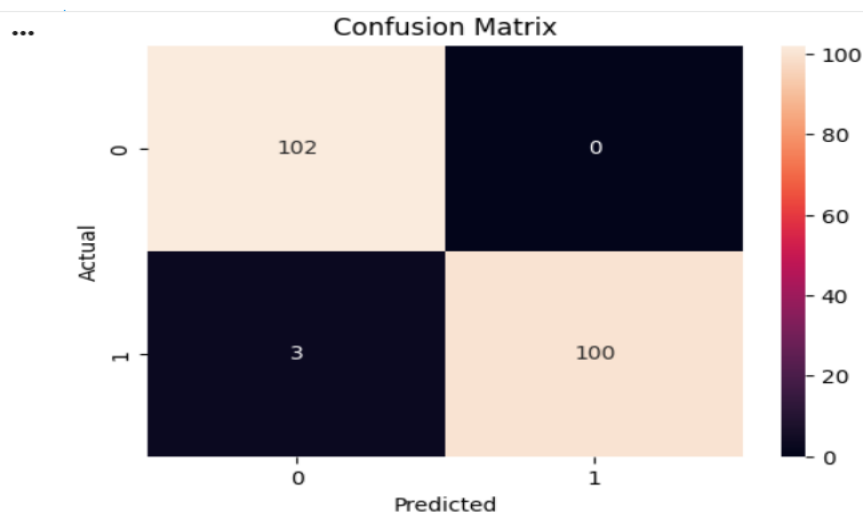
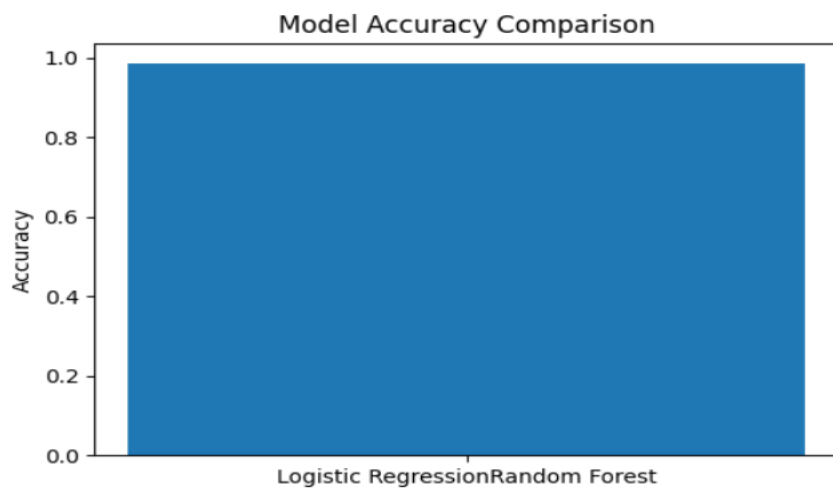
0.7951

Random Forest Accuracy:

0.9854

ROC-AUC Score:

1.0



Project Completed Successfully!

ROC-AUC Analysis

ROC-AUC was used to measure the model's ability to distinguish between patients with and without heart disease. A high ROC-AUC score indicated strong classification performance and model reliability.

Conclusion

This project successfully developed a heart disease prediction system using machine learning techniques. Random Forest outperformed Logistic Regression and demonstrated strong predictive capability. The results show that machine learning can effectively support healthcare professionals in disease prediction and risk assessment.

Future Scope

- XGBoost Implementation
- Hyperparameter Optimization
- Real-Time Clinical Data Integration
- Deployment as a Web Application

References

1. UCI Machine Learning Repository
2. Scikit-Learn Documentation
3. Kaggle Heart Disease Dataset